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Editorial

When AI Writes, Reviews, and Reads

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■ WHAT REMAINS UNIQUELY HUMAN IN SCIENTIFIC PUBLISHING?

In the past two years, research communities have seen a burst of activity in developing and testing artificial intelligence (AI) tools across specialized scientific disciplines. These advances have enabled new ways of composing, editing, and critiquing manuscripts and summarizing published work with minimal efforts. It is only a matter of time when these AI tools will be able to compose a manuscript based on the data input.

This Editorial takes a cautious but upfront look at the potential impact of AI on scholarly publishing. The question is not whether AI systems will be used in scholarly communication, but how their integration may reshape a human-centered model of authorship, peer review, and readership in ways that warrant careful scrutiny.

AI as an Author

AI systems trained in specific scientific subdisciplines, including chemistry, can now provide background information on a given topic along with relevant citations. On the one hand, such tools are valuable for gaining an overview of unfamiliar topics and identifying key references. On the other hand, they can also be used to compose the introduction for a manuscript or even review articles by gathering information across subtopics. As a result, we are beginning to see an increase in AI-generated or AI-assisted content in published review articles.[1] This trend raises important ethical questions: How should AI assistance be incorporated transparently into scholarly publications? What constitutes appropriate acknowledgment of AI-assisted contributions?[2] According to the AI Best Practices and Policies at ACS Publications, "AI tools cannot meet the requirements for authorship as they cannot take responsibility and accountability for the published work. As such, AI tools should not be included in the authorship list. Instead, AI tools should be acknowledged and mentioned transparently."[3]

AI-generated content is already beginning to appear in conference abstracts and scientific articles.[1,4] The emergence of platforms such as "aiXiv," which mirrors the human-administered "arXiv", now accept preprints authored by "AI scientists" and "robot scientists".[5] Although the proportion of AI-generated content in the published literature remains relatively small, it is likely to grow substantially in the coming years, and journals will face practical challenges. Should manuscripts containing AI-generated content become common, new frameworks will be required to evaluate consistently such submissions while preserving standards of originality, rigor and trust.

AI as a Reviewer

The increasing ease of composing scientific manuscripts is likely to lead to a surge in submissions, placing unprecedented strain on journal editorial systems. At the same time, the growing difficulty of recruiting qualified experts for peer review may make it tempting to employ AI-based tools to prescreen submissions or to assist with the review process.

AI-generated reviews typically analyze textual content, and provide general, surface-level feedback. They may be effective at identifying inconsistencies in presentation, missing elements, or deviations from expected structure. However, they cannot check the accuracy of data presentation in figures, evaluate the validity of experimental data, or critically assess whether a result constitutes a meaningful scientific advance. Such reviews primarily respond to prompted questions and tend to reflect consensus viewpoints, potentially overriding minority perspectives or failing to recognize truly disruptive or breakthrough research.[6,7] A recent survey suggests that up to 50% of submitted reviews now contain some level of AI-generated content.[8,9] Despite these shortcomings, AI-assisted review processes may reduce the burden on human reviewers and provide a preliminary screening for manuscript evaluation.

It is important to note, however, that many publishers currently prohibit the use of AI in peer review. For example, ACS Publications strictly disallows AI-generated reviews, and at *ACS Energy Letters*, reviews found to contain AI-generated content are rescinded. Human accountability and confidentiality are central to peer review; any use of tools should preserve both.[10] This policy underscores the continued importance of human judgment, expertise, and accountability in the peer-review process.

AI as a Reader

With the growing flood of papers published each day, it will become nearly impossible for readers to comprehensively track the literature. In this context, AI tools will likely to become indispensable, as they could rapidly summarize or generate structured reports from large sets of published articles. Such capabilities are known to reduce the burden of reading papers in full by allowing AI to scan the literature and highlight key findings and trends. As a result, detailed manual reading of

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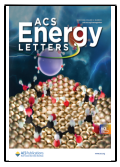

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published articles may become increasingly rare, with researchers relying more heavily on AI-assisted synthesis to navigate the expanding scientific record.

AI Driven Publishing Cycle

As AI tools become increasingly integrated into research workflows, they offer a convenient means to accelerate research output and navigate an increasingly competitive research landscape. If widely adopted, one can envision an AI-driven publication cycle (Figure 1) in which manuscripts are composed with AI assistance, reviewed by AI tools, and ultimately read and summarized by AI platforms. In such a scenario, an inevitable and essential question arises: *What is the role of humans in scientific publication?*

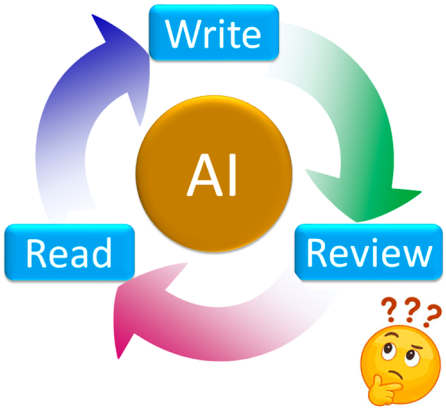

Figure 1. AI driven cycle of composing, reviewing and reading a manuscript.

The long-term success and credibility of research will therefore depend not merely on the adoption of AI, but on its responsible use, one that emphasizes the "**I**" (intelligence, insight, and integrity) rather than the "**A**" (artificial, automation and acceleration).[11]

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Notes

Views expressed in this editorial are those of the author and not necessarily the views of the ACS.

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